

Problem 69 (Hungary-Israel Team Competition 2001, Problem 1). The edges of a complete graph on n vertices, where $n \geq 3$, are coloured with n colours, and every colour appears at least once. Prove that one can find a triangle whose sides are coloured with 3 different colours.

Solution. Use induction on n . The case $n = 3$ is trivial. Assume that the claim holds for $n - 1$. Now consider a complete graph K_n on n vertices and a vertex a in K_n . Deleting a all its incident edges, we have a complete graph on $n - 1$ vertices. If this graph is coloured with at least $n - 1$ colours, then by the induction assumption, we are done. Otherwise it is coloured with at most $n - 2$ colours, say $1, 2, \dots, n - 2$. Now, since the remaining 2 colours, say $n - 1$ and n , must appear in K_n , there must exist 2 distinct vertices $b, c \neq a$ such that the edge ab is coloured with $n - 1$ and the edge ac is coloured with n . But the edge bc is not coloured with either $n - 1$ or n . Therefore the vertices a, b, c form a triangle whose sides are coloured with 3 different colours. $\square$

Problem 70 (Hungary-Israel Team Competition 2001, Problem 2). If the number of edges of a simple undirected graph on n vertices is at least $\frac{n^2}{4} + 2$, where $n \geq 5$, show that there exist 2 triangles which have exactly one common vertex.

Solution. Let G_n be a simple undirected graph on n vertices. We will use induction on n .

For $n = 5$, we have

$$e(G_5) \geq \frac{5^2}{4} + 2$$

and so $e(G_5) \geq 9$. Hence there is at most 2 vertices in G_5 that are not connected by an edge. Let v, w, x, y, z be the vertices and suppose that v, w are not connected by an edge. Then v, z, x and w, y, x are 2 triangles with a common vertex.

Assume that the claim holds for $n - 1$. Now consider G_n with $e(G_n) \geq \frac{n^2}{4} + 2$. If there is a vertex v in G_n with $d(v) < \frac{n}{2}$, then deleting this vertex and all its incident edges, we have a graph whose number of edges is at least

$$\frac{n^2}{4} + 2 - \frac{n}{2} + 1 \geq \frac{(n-1)^2}{4} + 2.$$

We apply the induction assumption to this graph and we are done. So we suppose that the degree of every vertex is at least $\frac{n}{2}$. Now since

$$\sum_{v \in V(G_n)} d(v) = 2e(G_n) \geq 2\left(\frac{n^2}{4} + 2\right) = \frac{n^2}{2} + 4 > \frac{n^2}{2}$$

there is at least one vertex v in G_n with $d(v) > \frac{n}{2}$, i.e.

$$d(v) \geq \left\lfloor \frac{n}{2} \right\rfloor + 1.$$

Let $v_1, \dots, v_r$ be the vertices adjacent to v , where

$$r \geq \left\lfloor \frac{n}{2} \right\rfloor + 1,$$

and let $w_1, \dots, w_s$ be the remaining vertices, where

$$s = n - r - 1 \leq \left\lceil \frac{n}{2} \right\rceil - 2.$$

Now $n \geq 6$ and so $r \geq 4$. For every $1 \leq i \leq r$, we have $d(v_i) \geq \frac{n}{2}$ and hence v_i is adjacent to at least one other vertex v_j .

Suppose that v_1 is adjacent to v_2 . If v_3 is adjacent to a vertex v_j with $j \neq 1, 2$, then the triangles v, v_1, v_2 and v, v_3, v_j have a common vertex. Hence we can assume that v_3 is adjacent to v_1 . For $4 \leq i \leq r$, if v_i is adjacent to a vertex v_j with $j \neq 1$, we are done. So suppose that v_1 is adjacent to all of the vertices $v_2, \dots, v_r$ and no other v_i, v_j are adjacent. It follows that each vertex v_i with $i \neq 2$ is adjacent to all of the vertices $u_1, \dots, u_s$ and that

$$s = \left\lceil \frac{n}{2} \right\rceil - 2.$$

Now consider $A = \{v_2, \dots, v_r\}$ and $B = \{v, v_1, u_1, \dots, u_s\}$ with $|A| = \left\lfloor \frac{n}{2} \right\rfloor$ and $|B| = \left\lceil \frac{n}{2} \right\rceil$. For every $x \in A$ and $y \in B$, x and y are adjacent. Moreover, v and v_1 are adjacent and no two elements of A are adjacent. Since $e(G_n) \geq \frac{n^2}{4} + 2$, then at least two more from B must be adjacent. The result follows. $\square$

Problem 71 (Hungary-Israel Team Competition 2001, Problem 3). If the number of edges of a simple undirected graph on n vertices is at least $\frac{n\sqrt{n}}{2} + \frac{n}{4}$, prove that the graph contains a cycle of length 4.

Solution. Suppose on the contrary that the claim does not hold, that is a graph G on n vertices satisfying the above inequality does not contain a cycle of length 4. We will count in two ways the number of elements of the set S given by

$$S = \{(u, \{v, w\}) : u, v, w \in V(G) \text{ where } u \text{ is adjacent to } v \text{ and to } w, \text{ with } v \neq w\}.$$

Summing over u , we have

$$|S| = \sum_{u \in V(G)} \binom{d(u)}{2}.$$

On the other hand, every pair $\{v, w\}$ has at most one common neighbour because there are no cycles of length 4. Hence $|S| \leq \binom{n}{2}$ and we conclude

$$\sum_{u \in V(G)} \binom{d(u)}{2} \leq \binom{n}{2},$$

or equivalently,

$$\sum_{u \in V(G)} d(u)^2 \leq n(n-1) + \sum_{u \in V(G)} d(u). \quad (*)$$

Using the Cauchy-Schwarz Inequality, we have

$$\left(\sum_{u \in V(G)} d(u) \right)^2 \leq n \sum_{u \in V(G)} d(u)^2.$$

Putting this in $(*)$ and using the fact that $\sum_{u \in V(G)} d(u) = 2e(G)$, we have

$$4e(G)^2 \leq n^2(n-1) + 2ne(G),$$

or equivalently,

$$e(G)^2 - \frac{n}{2}e(G) - \frac{n^2(n-1)}{4} \leq 0.$$

Solving this, we obtain

$$e(G) \leq \frac{n\sqrt{4n-3}}{4} + \frac{n}{4}$$

which is clearly strictly less than $\frac{n\sqrt{n}}{2} + \frac{n}{4}$. This yields the desired contradiction, and we are done. $\square$

Problem 72 (Hungary-Israel Team Competition 2001, Problem 4). Prove the following statements:

- (a) If a simple undirected graph on n vertices does not contain a complete bipartite graph with 2 vertices on one vertex class and 3 vertices on the other vertex class, then its number of edges is at most $\frac{n\sqrt{n}}{\sqrt{2}} + n$.
- (b) Given $n \geq 16$ distinct points $P_1, P_2, \dots, P_n$ in the plane, then at most $n\sqrt{n}$ of the segments $P_i P_j$ have unit length.

Solution. Let G be simple undirected graph on n vertices not containing a $K_{2,3}$.

(a) We will prove a stronger result, namely,

$$e(G) \leq \frac{n\sqrt{n}}{\sqrt{2}} + \frac{n}{4}.$$

Now we use the same set S in the previous problem. This time, however, we have $|S| \leq 2\binom{n}{2}$ since every $\{v, w\}$ has at most two common neighbours. Then, again, by the Cauchy-Schwarz Inequality and the fact that $\sum_{u \in V(G)} d(u) = 2e(G)$, we have

$$-e(G) + \frac{2}{n}e(G)^2 \leq -e(G) + \frac{1}{2} \sum_{u \in V(G)} d(u)^2 \leq n(n-1),$$

or equivalently,

$$2e(G)^2 - ne(G) + n^2(n-1) \leq 0.$$

Solving this, we have

$$e(G) \leq \frac{n\sqrt{8n-7}}{4} + \frac{n}{4}$$

which is clearly strictly less than $\frac{n\sqrt{n}}{\sqrt{2}} + \frac{n}{4}$, and we are done. $\square$

(b) Here, we also prove a stronger result: the claim holds for every positive integer value of n .

Consider a graph G with the points $P_1, \dots, P_n$ as vertices and two vertices are adjacent if and only if $P_i P_j$ has unit length. Now observe that given two points P, Q in the plane, there are exactly two points in the plane that have a unit distance with P and Q . Hence G does not contain a $K_{2,3}$ and by (a),

$$e(G) \leq \frac{n\sqrt{n}}{\sqrt{2}} + \frac{n}{4} \leq n\sqrt{n}.$$

The last inequality holds for every $n \geq 2$. The case $n = 1$ is trivial. $\square$